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RALPH O'CONNOR SUSTAINABLE ENERGY INSTITUTE

Beyond Market Signals:
Supply-Chain Constraints in Grid Expansion

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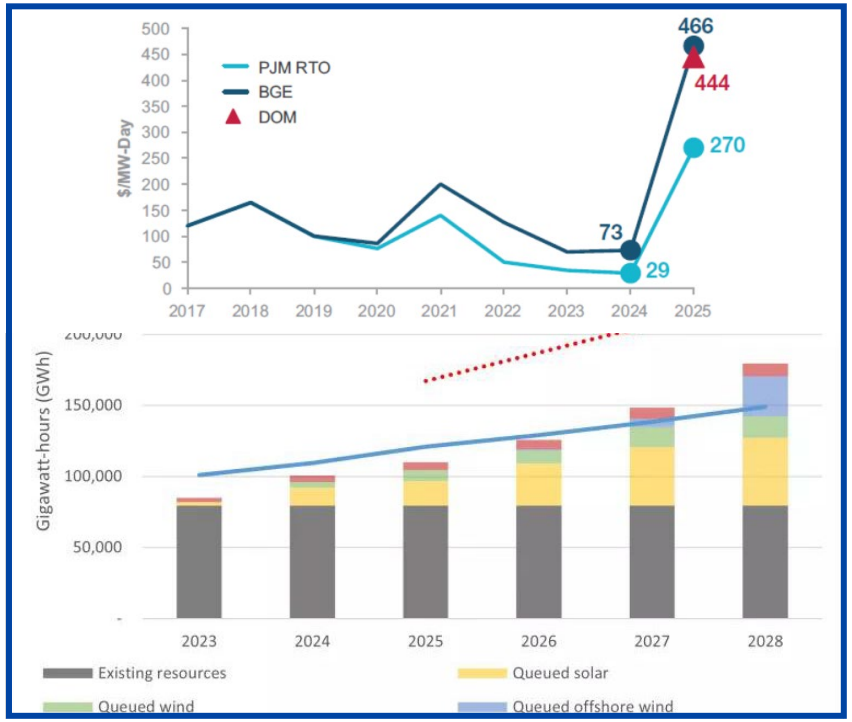
Two stories reveal the mismatch between plans and supply chains

- Story 1 — New York Wind: The Broken Promise
  - **2021:** Contracts for major offshore wind farms (e.g., Empire Wind 2) promised power for millions [1].
  - Contracts locked in, market signals clear, OREC incentives secured [2], developers ready to build.
  - **By 2024:** Inflation, **turbine and cable shortages**, and supply chain bottlenecks halted projects [3].
  - Contracts collapsed — markets pointed forward, incentives aligned, but **supply chains held back**.

- Story 2 — PJM Market: The Price Shock
  - **2024:** PJM capacity auction prices **spiked up to 800%** in some zones [4].
  - Consumers and businesses faced billions in extra costs**.**
  - Projects stalled in queues, **transformers, materials**, and financing all **in short supply** [5].
  - Market signals were strong, but **physical bottlenecks** blocked expansion.



[1] https://www.nyserda.ny.gov/All-Programs/Offshore-Wind/Focus-Areas/NY-Offshore-Wind-Projects
[2] https://www.nyserda.ny.gov/All-Programs/Offshore-Wind/Focus-Areas/Offshore-Wind-Solicitations/2020-Solicitation
[3] https://patch.com/new-jersey/longbranch/empire-wind-2-turbines-canceled-would-have-been-19-miles-nj
[4] https://www.reuters.com/business/energy/pjm-power-auction-results-yield-sharply-higher-prices-2024-07-31
[5] https://opc-dc.gov/wp-content/uploads/2025/05/PJM-Capacity-Market-Report-FINAL-OPC-Synapse.pdf

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Two stories reveal the mismatch between plans and supply chains

Incentives ≠ Deployment!

Supply chain, not contracts, dictated what was feasible.

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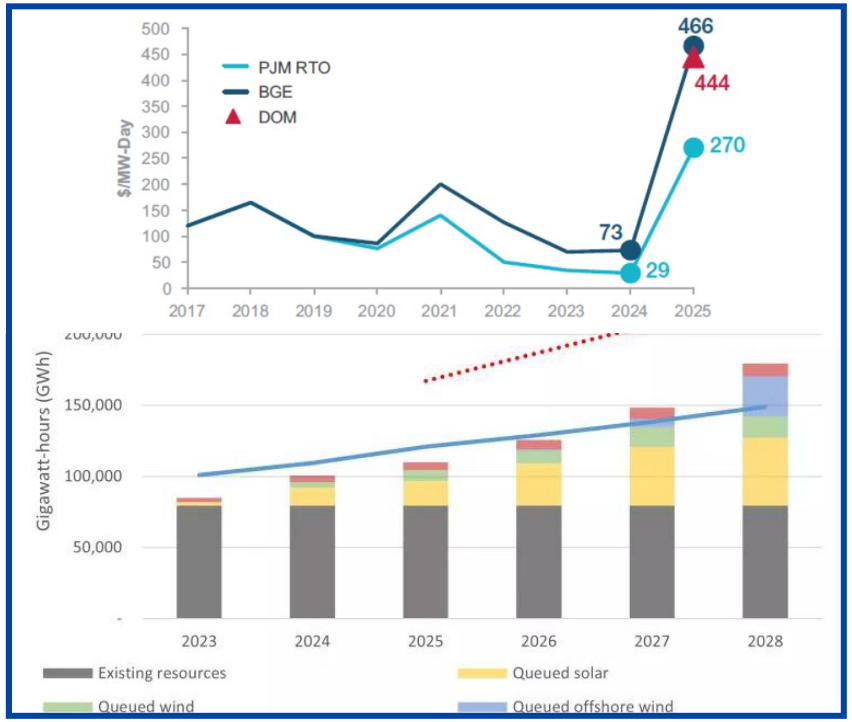

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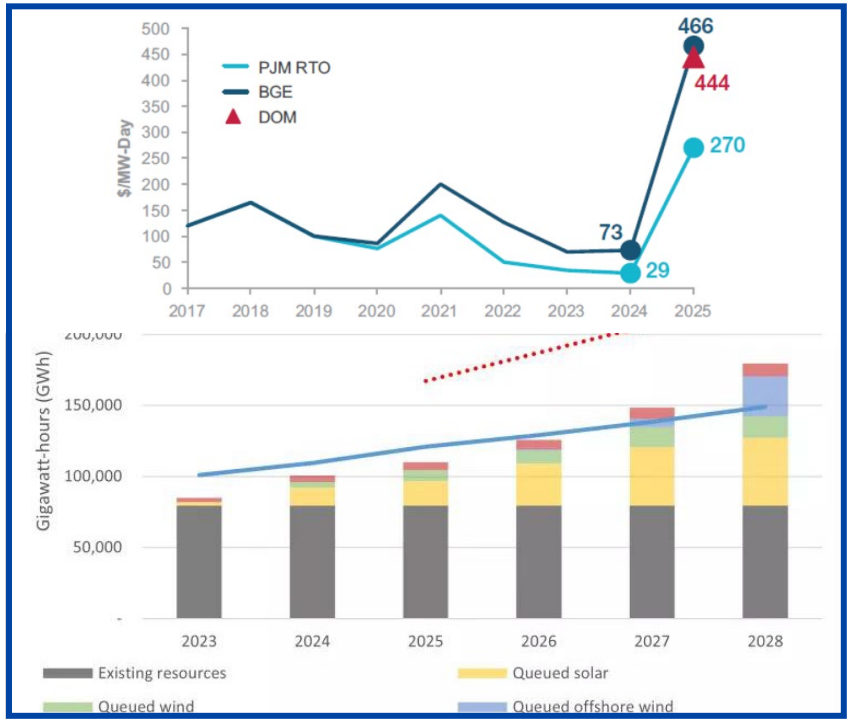
Incentives ≠ Deployment!

Supply chain, not contracts, dictated what was feasible.

Prices ≠ Projects!

Supply chain, not market signals, determined what could proceed.



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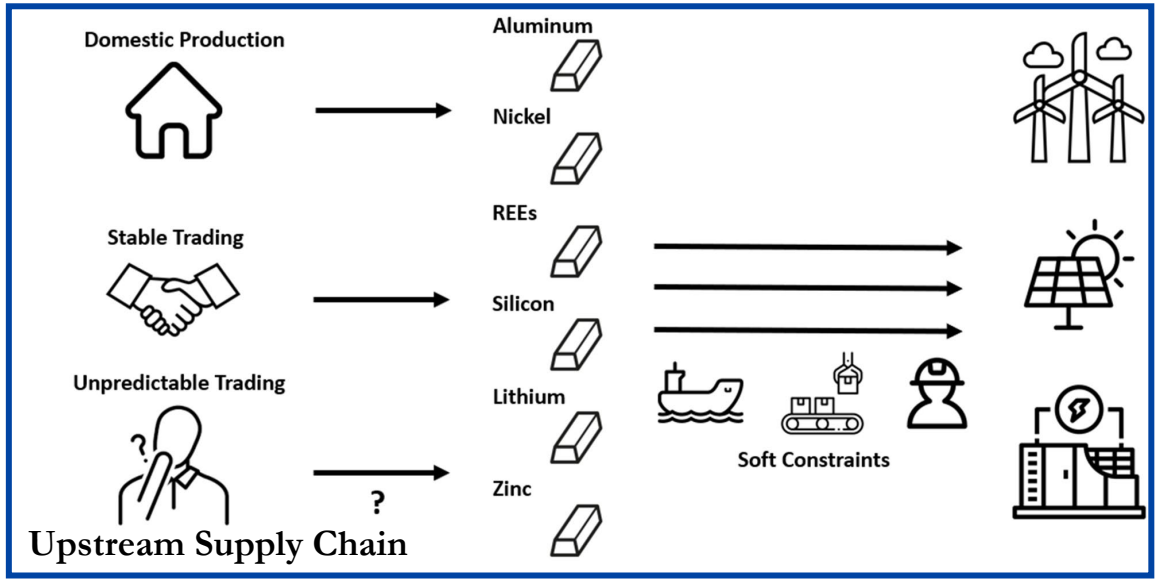
Supply chains for generation resources involve significant critical materials [8, 9]

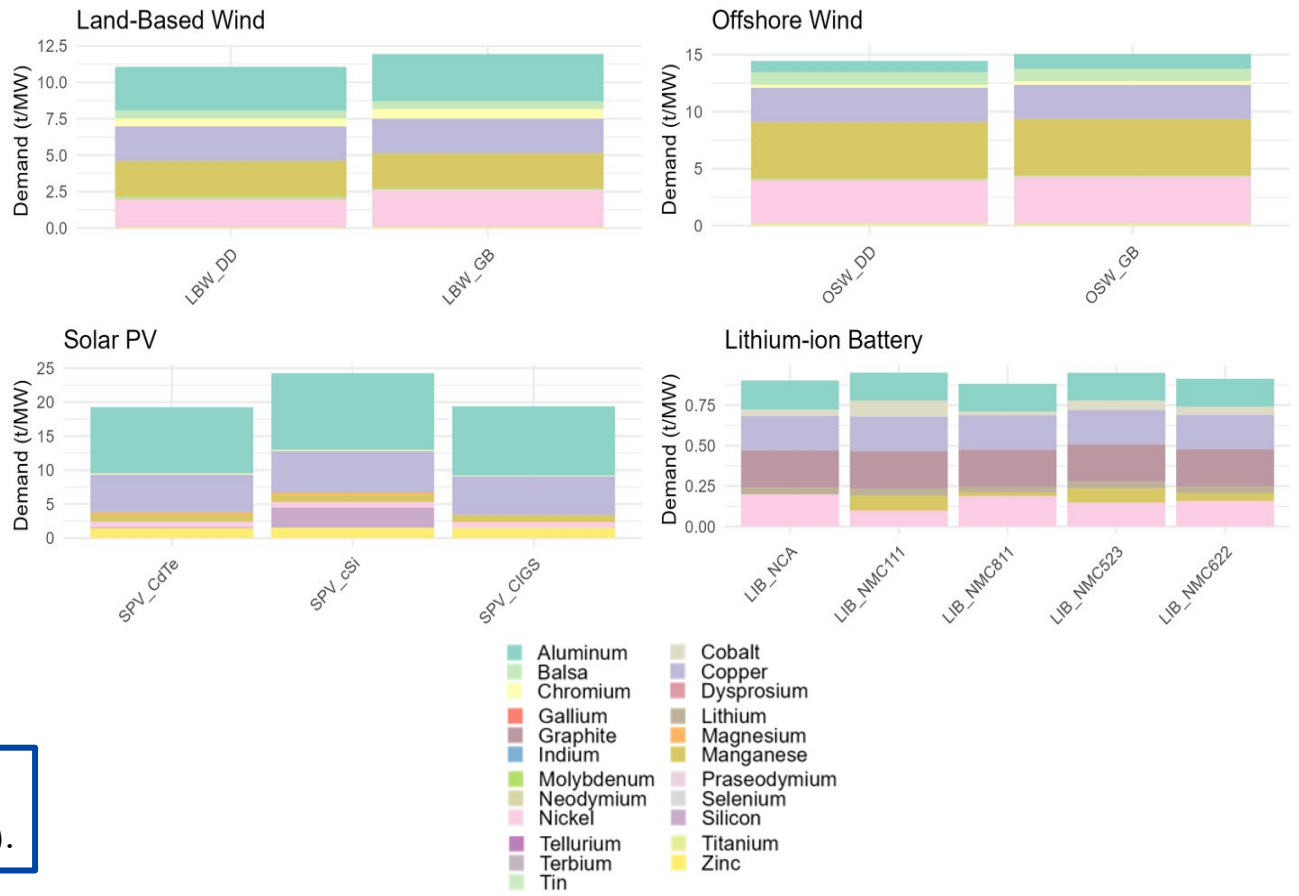


Materials → Components → Products

26 Critical materials from USGS and DOE critical lists [10, 11]; Sources categorized as: Stable vs. Unpredictable

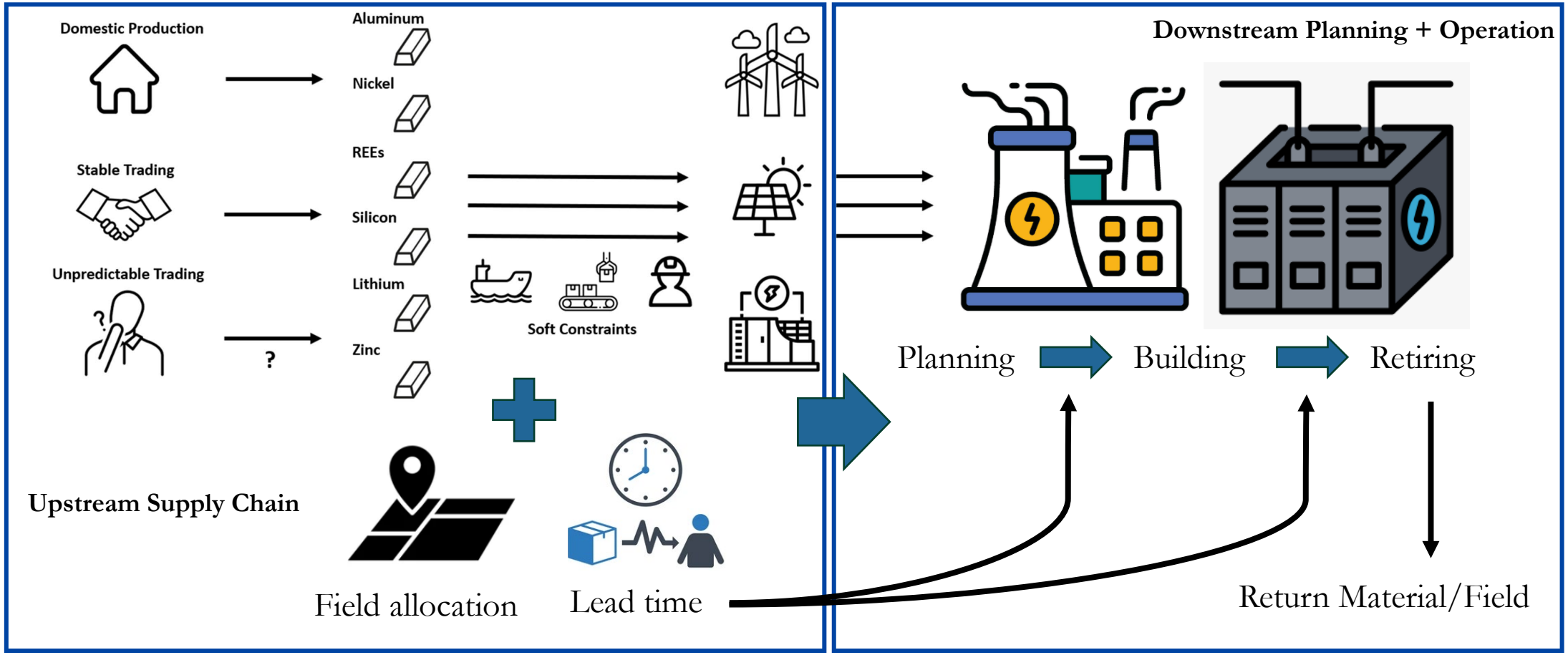
Established **our** dataset on material intensity (e.g., tons/MW).

Wind: Onshore & Offshore (Gearbox, Direct-drive)
Solar PV: c-Si, CIGS, CdTe
Batteries: Li-ion (NCA, NMC-111/523/622/811, LFP)



[8] Yao, B., & Dvorkin, Y. (2025). A decision framework for clean energy supply chain optimization. In Proceedings of the IEEE Power & Energy Society General Meeting (PESGM). IEEE.
[9] Yao, B., Jeong, H., Mehrtash, M., Allan, B., Ockerman, D., & Dvorkin, Y. (2025). Understanding supply chain constraints for the US clean energy transition. npj Clean Energy, 1(1), 9.
[10] U.S. Geological Survey, 2024, Mineral commodity summaries 2024: U.S. Geological Survey, 212 p., https://doi.org/10.3133/mcs2024
[11] Bauer, D. J., Khazdozian, H., et al. (2023). 2023 Critical Materials Strategy (INL/RPT-23-72323-Rev.001). Idaho National Laboratory.

No unified model linking supply chains to capacity expansion planning



Integrating upstream supply chains into generation expansion planning

obj.

$$\min \sum_y C_y = \sum_y \left(C_y^{\text{in}} + C_y^{\text{op}} + C_y^{\text{pe}}\right) \quad (1a)$$

$$C_y^{\text{in}} = \sum_{g \in \tilde{\mathcal{G}}} AC_{gy}^{\text{I}} \cdot \overline{P}_g^{\text{G}} \cdot d_{gy} \quad (1b)$$

$$C_y^{\text{op}} = \sum_g C_g^{\text{F}} \cdot \overline{P}_g^{\text{G}} \cdot o_{gy} + \sum_{g \in \mathcal{G}^{\text{th}} \cup \mathcal{G}^{\text{rn}}} \sum_t C_g^{\text{V}} \cdot N_{ty} \cdot \sum_h p_{gthy} + \sum_{g \in \mathcal{G}^{\text{st}}} \sum_t C_g^{\text{V}} \cdot N_{ty} \cdot \sum_h (c_{gthy} + dc_{gthy}) \quad (1c)$$

$$C_y^{\text{pe}} = \sum_t N_{ty} \cdot P^{\text{VOLL}} \cdot \sum_h p_{ithy}^{\text{LS}} + P^{\text{RM}} \cdot p_y^{\text{RM}} + P^{\text{RPS}} \cdot e_{ky}^{\text{RPS}} \quad (1d)$$

1.

$$u_{my} \geq \sum_c v_{cy} \cdot D_{mc}^{\text{CO}}, \quad \forall m, y \quad (2a)$$

$$v_{cy} \geq \sum_p w_{py} \cdot D_{cp}^{\text{PR}}, \quad \forall c, y \quad (2b)$$

$$u_{my} \leq \overline{M}_{my} + \sum_{g \in \mathcal{G}} R_{mg}^{\text{RM}} \cdot \overline{P}_g^{\text{G}} \cdot r_{gy} + s_{my}, \quad \forall m, y \quad (2c)$$

$$s_{my} = s_{m(y-1)} + \overline{M}_{m(y-1)} - u_{m(y-1)} + \sum_{g \in \mathcal{G}} R_{mg}^{\text{RM}} \cdot \overline{P}_g^{\text{G}} \cdot r_{g(y-1)}, \quad \forall m, y \quad (2d)$$

2.

$$\sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} \cdot d_{gy} \leq \sum_{p \in \mathcal{P}^k} w_{py}, \quad \forall k, y \quad (2e)$$

$$\frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot d_{gy}}{R_k^{\text{CAP}}} \leq f_{iy}^k, \quad \forall i, y, k \quad (2f)$$

3.

$$f_{iy}^k = f_{i(y-1)}^k + \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot r_{gy}}{R_k^{\text{CAP}}} - \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot d_{g(y-1)}}{R_k^{\text{CAP}}}, \quad \forall i, y > 1, k \quad (2g)$$

$$f_{i1}^k = A_i^k + \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot r_{g1}}{R_k^{\text{CAP}}}, \quad \forall i, k \quad (2h)$$

4.

$$\sum_{y'' \leq y - T_g^{\text{LEAD}}} d_{gy''} = \sum_{y' \leq y} b_{gy'}, \quad \forall g \in \tilde{\mathcal{G}}, \ y \quad (2i)$$

$$b_{gy} = 0, \quad \forall g \in \bar{\mathcal{G}}, \ y \quad (2j)$$

$$\sum_{y'' \leq y - T_g^{\text{LT}}} b_{gy''} = \sum_{y' \leq y} r_{gy'}, \quad \forall g \in \tilde{\mathcal{G}}, \ y \quad (2k)$$

$$r_{gy} = 1 \text{ if } y = T_g^{\text{RT}}, \ 0 \text{ if } y < T_g^{\text{RT}}, \quad \forall g \in \tilde{\mathcal{G}}, \ \forall y \quad (2l)$$

Objective =
Investment + Operations + Penalties

Upstream supply chain constraints

1. Materials → components → products.
2. Field availability.
3. Recovery from retired units.
4. Limits by production & lead times.

5.

$$\sum_{g \in \mathcal{G}_i \cap (\mathcal{G}^{\text{th}} \cup \mathcal{G}^{\text{rn}})} p_{gthy} + \sum_{g \in \mathcal{G}_i \cap \mathcal{G}^{\text{st}}} (dc_{gthy} - c_{gthy}) - \sum_{l \in \mathcal{LS}_i} q_{lthy} + \sum_{l \in \mathcal{LR}_i} q_{lthy} = L_{ithy} - p_{ithy}^{\text{LS}}, \quad \forall i, t, h, y \quad (3a)$$

6.

$$0 \leq p_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{th}}, \ y, t, h \quad (3b)$$

$$0 \leq p_{gthy} \leq F_{igthy}^{\text{GEN}} \cdot \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}_i \cap \mathcal{G}^{\text{rn}}, \ i, y, t, h \quad (3c)$$

7.

$$o_{gy} = o_{g(y-1)} + b_{gy} - r_{gy}, \quad \forall g \in \mathcal{G}, \ y \geq 1 \quad (3d)$$

$$o_{g0} = 1 \text{ if } g \in \bar{\mathcal{G}}, \ 0 \text{ otherwise}, \quad \forall g \in \mathcal{G} \quad (3e)$$

$$-\overline{P}_l^{\text{L}} \leq q_{lthy} \leq \overline{P}_l^{\text{L}}, \quad \forall l, y, t, h \quad (3f)$$

$$\sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} \cdot F_{ky}^{\text{ELCC}} \cdot o_{gy} + p_y^{\text{RM}} \geq (1 + R_y^{\text{RM}}) \cdot \overline{L}_y, \quad \forall y, k \quad (3g)$$

$$0 \leq p_{ithy}^{\text{LS}} \leq L_{ithy}, \quad \forall i, y, t, h \quad (3h)$$

8.

$$\sum_{g \in \mathcal{G}^k} \sum_t \sum_h N_{ty} \cdot p_{gthy} + e_{ky}^{\text{RPS}} \geq R_{ky}^{\text{RPS}} \cdot \sum_t \sum_h \sum_i N_{ty} \cdot L_{ithy}, \quad \forall k, y \quad (3i)$$

9.

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Downstream operation constraints

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Integrating upstream supply chains into generation expansion planning

- **Decisions (annual): where/when/what** to build; **product mix** (Solar/Wind/Battery variants).
- **Operations (hourly, rep. days):** dispatch, storage charge/discharge, flows, load shedding.
- **Upstream layers:** materials → components → products; lead time; stocks & recovery from retirements; field availability.

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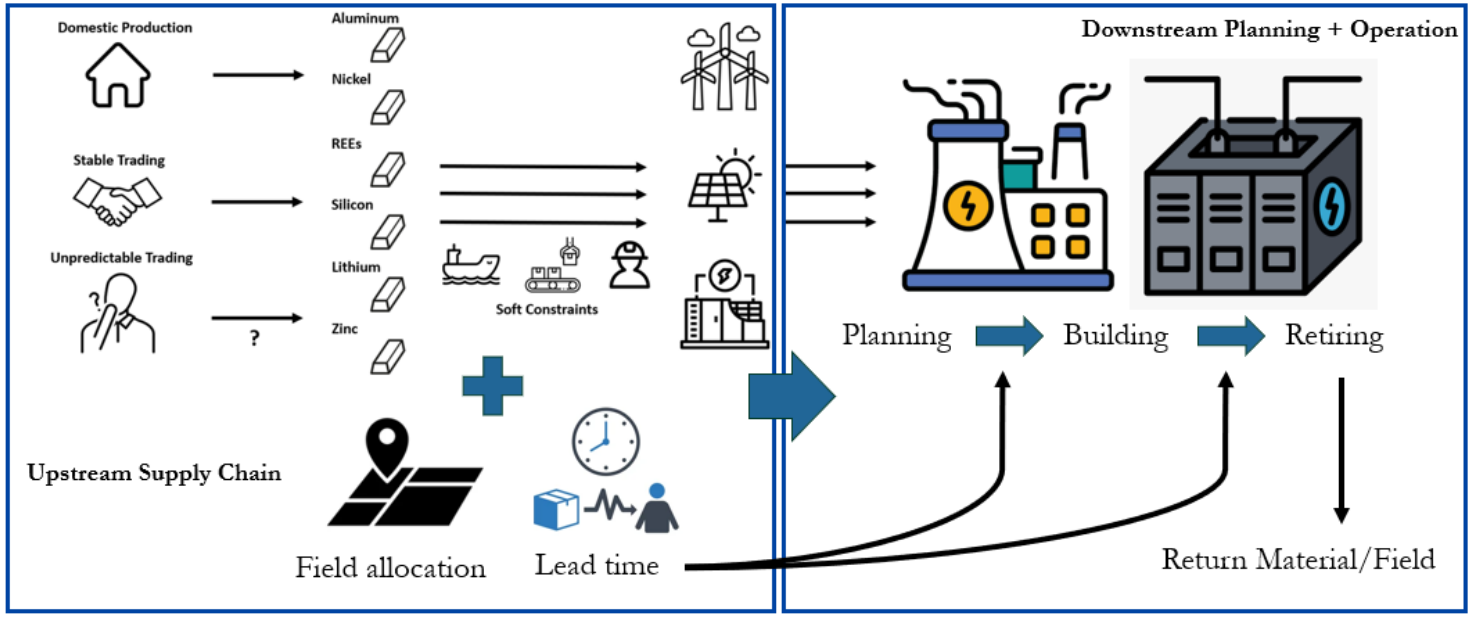
Objective = Investment + Operations + Penalties
Build new capacity, run the system, cover shortfalls.



1. Materials → components → products.
2. Limits by production & lead times.
3. Field availability.
4. Recovery from retired units.

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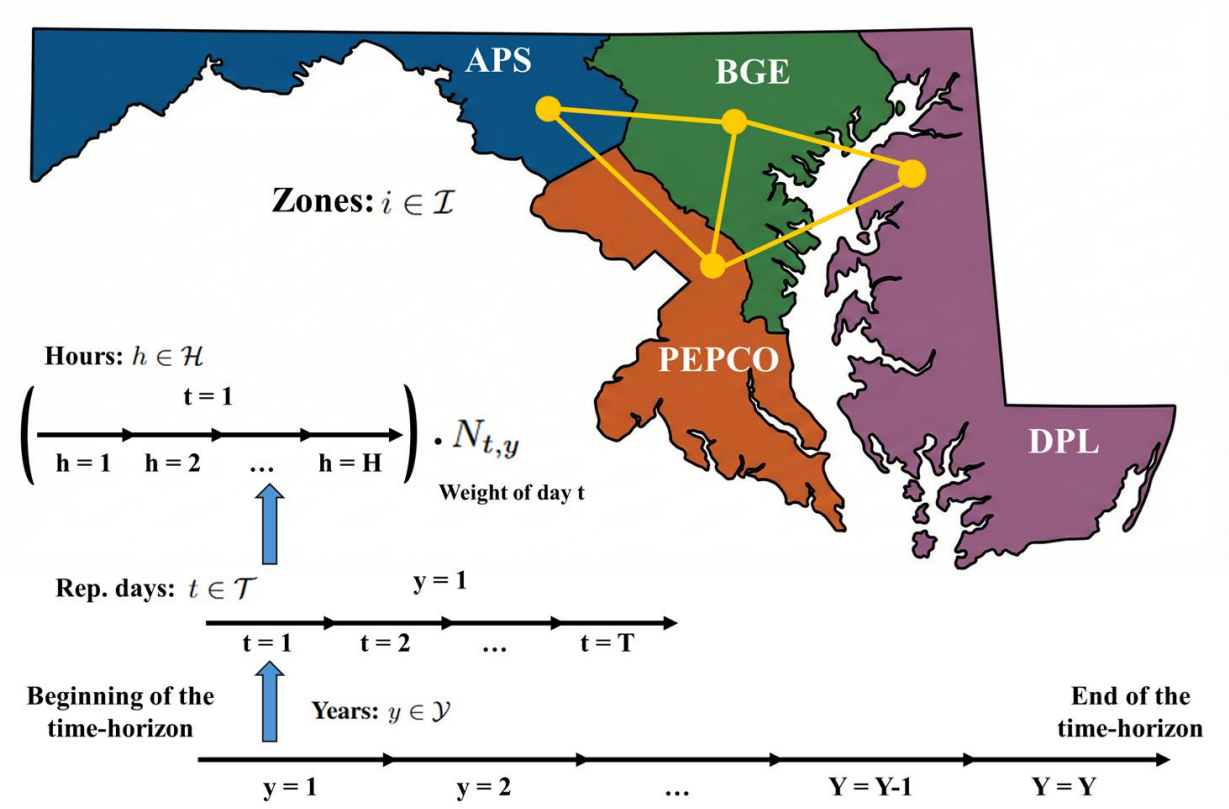
Maryland case study: supply chains shape generation planning



Maryland spatial and temporal representation

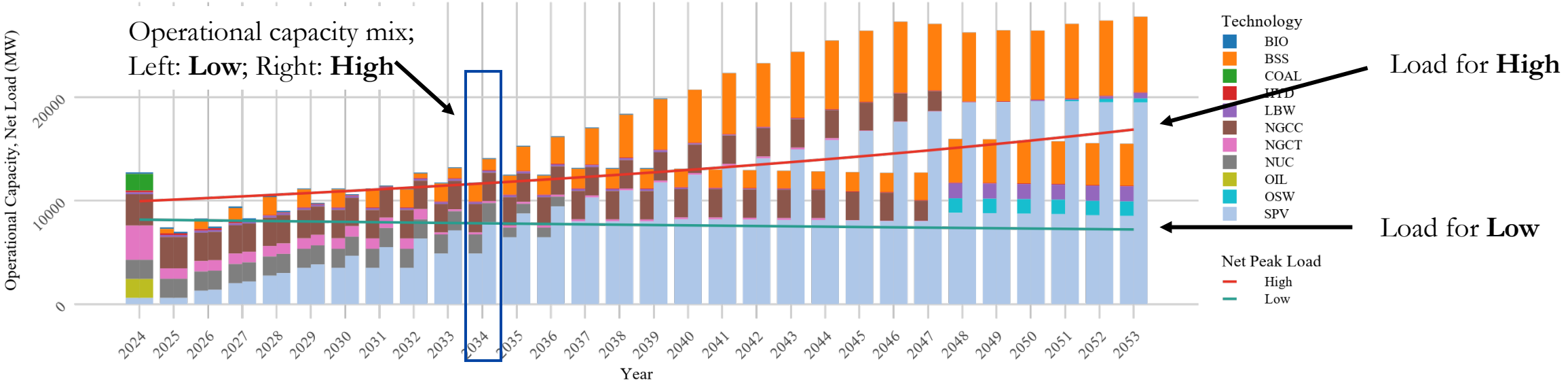
Scenarios:

- **Low**: no data centers, low electrification.
- **High**: with data centers, high electrification.
- Sub-Scenarios:
  - **w/o SC** (relaxed): **unlimited** materials, no lead times.
  - **lim. SC** (constrained): **domestic/allied materials only**, with lead times & land limits.

Replicate traditional generation planning.

Stricter material limits.

Maryland case study: supply chains shape generation planning

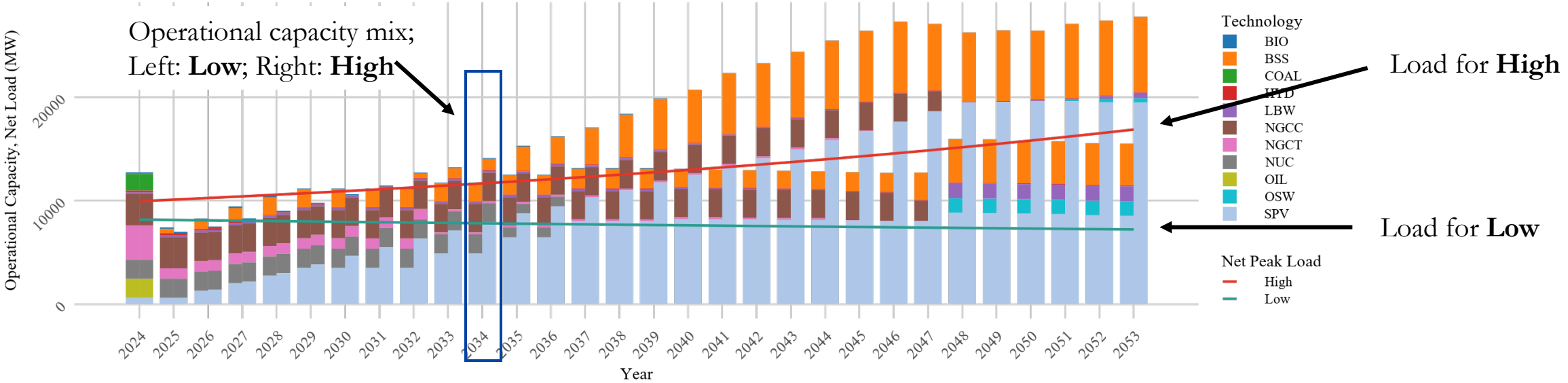


Operational capacity over time for baseline Low and High scenarios. Stacked bars: tech-specific capacity (Low baseline = left); Lines: net peak load (system demand)

- Low baseline (2053): 4.1 GW BSS, 1.5 GW LBW, 1.4 GW OSW, 8.5 GW SPV → diversified mix.
- High baseline (2053): 7.3 GW BSS, 0.5 GW LBW, 0.4 GW OSW, 19.5 GW SPV → solar-dominant.

Rising demand + Tighter materials = **Lower flexibility & feasibility**

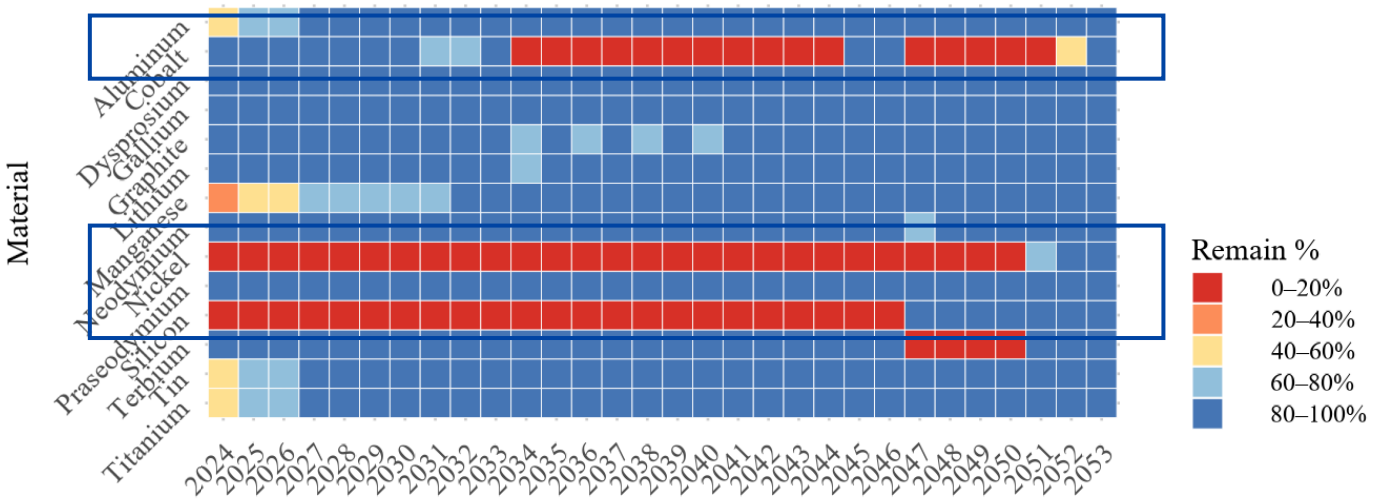
Maryland case study: supply chains shape generation planning



Operational capacity over time for baseline Low and High scenarios. Stacked bars: tech-specific capacity (Low baseline = left); Lines: net peak load (system demand)

- Low baseline (2053): 4.1 GW BSS, 1.5 GW LBW, 1.4 GW OSW, 8.5 GW SPV → diversified mix.
- High baseline (2053): 7.3 GW BSS, 0.5 GW LBW, 0.4 GW OSW, 19.5 GW SPV → solar-dominant.

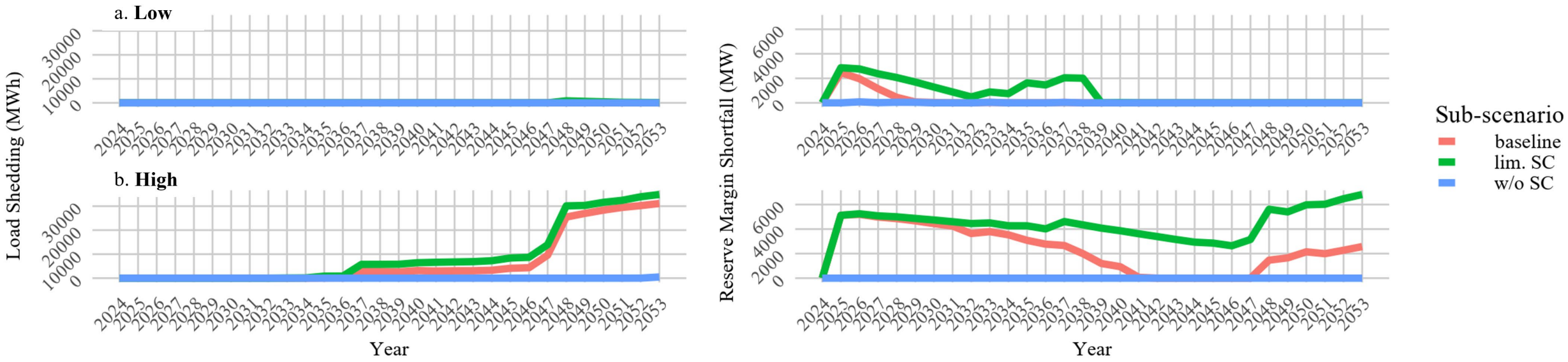
Why this shift?



Nickel and cobalt scarcity blocks wind deployment.

JOHNS HOPKINS
RALPH O'CONNOR SUSTAINABLE ENERGY INSTITUTE

Reliability threatened by upstream supply chains



- **Blue lines:** no supply chain limits → just-in-time builds, no physical limits → no load shedding, no reserve margin shortfall.
- **Red lines:** with supply chain limits → insufficient manufacturing capacity → gas plant retirements cause load shedding and reserve margin violations.
- **Green lines:** stricter material limits → fewer resources → greater reliability risk.

Key takeaways

- Ignoring supply chains underestimates both cost and reliability risks.
- Just-in-time planning is infeasible → requires earlier, strategic investments and lifetime extensions.
- Constraints can erase cost advantages (no materials, no MW) → diversification across technologies is essential.
- Supply chains are emerging as a key determinant of grid and system feasibility.